



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 10-402-1
18-01-2022

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SPECIFICATION

FOR

LONG-ROD TENSION-TYPE

POLYMER INSULATOR

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1. SCOPE

This specification describes the minimum technical requirements for design, engineering, manufacturing, testing, inspection, packing, and performance of outdoor long rod tension-type polymer insulators. The required insulators are intended to be installed in the overhead lines of the medium voltage distribution networks (12 kV and 24 kV) of the Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification, the offered polymer insulators should be designed, manufactured and tested according to the latest editions of the relevant applicable standards given in Table (1). In case of any deviation from the listed standard, it should be indicated in the deviations list submitted by the supplier.

Table (1)

Standard No.	Description
IEC 60120	Ball and socket couplings of string insulator units – Dimensions
IEC 60815	Selection and dimensioning of high-voltage insulators intended for use in polluted conditions
IEC 60383	Insulators for overhead lines with a nominal voltage above 1000 V
IEC 61109	Insulators for overhead lines - Composite suspension and tension insulators for A.C. systems with a nominal voltage greater than 1000 V - Definitions, test methods and acceptance criteria
IEC 62217	Polymeric HV insulators for indoor and outdoor use - General definitions, test methods and acceptance criteria
IEC 61952	Insulators for overhead lines - Composite line post insulators for A.C. systems with a nominal voltage greater than 1000 V - Definitions, test methods and acceptance criteria
IEC 60273	Characteristic of indoor and outdoor post insulators for systems with nominal voltages greater than 1000 V
IEC TR 62039	Selection guide for polymeric materials for outdoor use under HV stress



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IEC 60437

Radio Interference Test on High-Voltage Insulators

3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required insulator should be guaranteed under the environmental conditions given in Table (2). Any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Ambient temperature	-5°C to +45°C (50°C as option according to ...EDC) Very intensive sunlight of 70 °C exposure temperature is expected.
Maximum relative humidity	Up to 100 %
Altitude	Up to 1000 m above sea-level
Pollution	Class D (heavy)
Average thunder storm	45 Days per annum.
Climatic condition	Moderately hot and humid tropical climate, conductive to rust and fungus growth.
Maximum annual rainfall	250 mm
Maximum wind velocity	55 km/hour
Solar energy radiation	$\geq 1100 \text{ W/m}^2$

4. SYSTEM CHARACTERISTICS

The required insulator should have a safe electrical insulation behavior for the electrical system given in Table (3).

Table (3)

Nominal voltage	11 and 22 kV
Highest system voltage	12 and 24 kV
Power frequency	50 Hz
Number of phases	Three-phase
System earthing	Solidly earthed



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5. TECHNICAL REQUIREMENTS

5.1 General:

- The long rod tension–type composite silicone rubber insulators shall meet all requirements of this specification in all respects and shall consist of a single piece of silicon rubber together with the core and its metal parts to form a rigid assembly.
- Manufacturer’s drawings shall show the outline of the insulators together with all pertinent dimensions. Any variation in these dimensions due to manufacturing tolerances shall be indicated.
- The tolerances on dimensions (e.g. diameter, length and creepage distance) shall be in accordance with IEC 61109. However, negative tolerance shall not be applicable to creepage distance. The tolerance on dimension d (in millimeters) is estimated as:
 $\pm (0.04d + 1.5)$ mm when $d \leq 300$ mm
 $\pm (0.025d + 6)$ mm when $d > 300$ mm with a maximum tolerance of ± 50 mm,
- The design shall be such that stresses due to expansion and contraction in any part of the insulator shall not lead to any deterioration. Precautions shall be taken to avoid any chemical reaction between insulator parts and fitting materials
- The insulators shall be suitable for overhead line conductors of cross-sectional area starting from 35 mm^2 up to 300 mm^2 .
- Long rod polymer insulators shall be designed to meet the quality, safety, reliability and capability to withstand a wide range of environmental conditions:
 - Core: The internal insulating part
 - Housing: The external insulating part.
 - Metal end fittings: For attaching hardware to support conductor.
- The insulator shall be radio interference free at operating voltage.



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- The insulators shall be made of best quality material free from flaws, cracks, voids, conducting impurities and chipped edges.
- The complete insulator with fittings shall not create excessive or localized corona at a voltage less than its rated voltage.
- The design of insulator and its fitting shall minimize local corona formation and discharges causing radio interferences.
- The electrical and mechanical characteristics for long rod polymer insulators shall be as given in Table (4).

Table (4)

1	Maximum system voltage (kV)	12	24
2	Operating voltage (kV)	11	22
3	Nominal Frequency (Hz)	50	
4	Wet power frequency withstand voltage (kV)	≥ 45	≥ 90
	Wet power frequency flashover voltage (kV)	≥ 50	≥ 100
5	Dry power frequency withstand voltage (kV)	≥ 70	≥ 110
	Dry power frequency flashover voltage (kV)	≥ 75	≥ 125
6	Impulse 1.2 / 50 μ s withstand voltage (kV)	≥ 115	≥ 170
7	Leakage distance not less than (mm)	480	960
8	Minimum breaking load (kN)	90	
9	Corona extinction voltage (kV)	9	18

5.2 Profile:

- The long rod polymer insulator shall have aerodynamic profile sheds with good hydrophobic and recovery properties. Additionally, it should be maintenance-free in accordance to IEC 60815-3.



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- The sheds shall have alternating profile of uniform spacing between sheds. The shed shape, spacing and inclination shall comply with IEC 60815-3 and shall be able to withstand the different types of pollution.

5.3 Construction:

- Long rod silicone rubber insulator must be manufactured to form one-piece unit. The complete insulator sets shall have minimum specified mechanical load (SML) 90 kN.

5.4 Core:

- The long rod composite silicone rubber insulator has a solid Fiber Reinforced Plastic (FRP-rod) core with silicone rubber housing. The diameter of core and housing should not be less than 23 mm.
- The FRP-rod core shall be epoxy resin reinforced with Electrical Corrosion Resistance (ECR), glass fibers and free of boron. It shall be resistant against acid attacks in order to prevent stress corrosion of the FRP-rod core.

5.5 Housing:

- The housing shall be made of composite silicone-elastomer-rubber compound with a thickness greater than 3.5 mm over the sheath. It shall be one-piece using compression or injection molding principle to cover the core.
- The base polymer shall be silicone rubber prior to the addition of reinforcing fillers.
- The housing shall be made of High Temperature Vulcanizing (HTV) silicone rubber material.
- The weather sheds shall be made of silicone elastomer compound molded as part of the sheath and shall be free from imperfections. The weather sheds should be the same silicone of the housing. It should protect the FRP-rod against environments influences, external pollution and humidity.



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- Sheds and sheath shall be produced in one molding step. Shed-sheath interface is unacceptable. The sheath must be homogeneous around the FRP-rod.
- Sheds shall be characterized by an aerodynamic profile of alternating sheds comply with the requirements of IEC 60815-3. The diameter of insulator sheds shall not exceed 160 mm.
- Based on 40 mm/kV, the creepage distance of the polymeric insulator shall be as follows:
 - For 12 kV insulators: the minimum creepage distance is 480 mm.
 - For 24 kV insulators: the minimum creepage distance is 960 mm
- The housing can be injected over the core in one shot. This provides a one- piece housing directly and chemically bonded to the core and the metal fittings. Alternatively, the housing can be compression molded onto the rod. These also result in a chemical bond to the core but the housing-to-end fitting seal has to be added.
- The mold lines must not exceed 1 mm in height.
- The sheath must be homogenous around fiber-glass rod.
- Insulator color should be light grey.

5.6 Metal end Fittings:

- End fitting shall be made of forged steel or cast iron. It shall be applied on to the corrosion resistant rod by a multi-point longitudinal crimping system to ensure uniformity and prevent the ingress of moisture. End fitting shall be compressed on the fiber glass rod. The interference, between fitting and housing shall be sealed tightly to avoid interface problem.
- The dimensions of end fittings of insulator shall be in accordance with the standard dimensions stated in IEC 60120.



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- The designated size of ball coupling should be size 16A. The socket could be size 16A or 16B in accordance to IEC 60120 standard.
- End fittings transmit the mechanical load to the core. Hardware of respective specified mechanical load should be hot-dip galvanized and Zinc coated with minimum 99.95 % purity of electrolytic high grade Zinc. The material used in fittings shall be corrosion resistant.
- Metal end fittings shall have smooth and uniform surfaces without any sharp edges or corners. It shall be free of cracks, flakes, slag, blow-holes shrinkages defects and localized porosity.
- They shall be connected to the rod by means of a controlled compression technique.
- As the main duty of the end fittings is the transfer of mechanical loads to the core, the fittings should be properly attached to the core by a coaxial or hexagonal compression process and should not damage the individual fibers or crack the core.
- The dimensions of end fittings of insulators shall be in accordance with the standard dimensions stated in IEC 60120.
- The finished surface shall be smooth with a good performance. The surface shall not crack or get chipped due to aging effect under normal and abnormal service conditions.
- Fittings should be effectively sealed to prevent moisture ingress. Effectiveness of sealing system must be supported by test documents.
- All surfaces of the metal parts shall be perfectly smooth without projecting points or irregularities, which may cause corona.



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6. PACKING and MARKING

- The insulators should be packed in suitable wooden crates, cases, or safe carton boxes to withstand shipment, loading, handling and storing.
- Each insulator shall bear a permanent marking in English language in accordance with IEC standard on the end of fitting.
- All the characters shall be legible, durable and permanently marked on one end fitting or on the uppermost shed of the insulator.
- The use of stickers/labels is not permitted.
- Each insulator shall bear the following information:
 - a) Breaking failing load.
 - b) Manufactures names
 - c) Year of manufacture-
 - d) Order number
 - e) Type and model of insulator
 - f) Rated voltage.
- Crate Marking (English and/or Arabic):
 - a) Nominal system voltage
 - b) Type and No. of insulators
 - c) Order number
 - d) Weight, kg.
 - e) Manufactures names and Name of the manufacturing country.



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7. TESTING and INSPECTION

- All tests shall be carried out by and at the expense of the supplier in accordance to the latest relevant standard.
- ...EDC reserves the right to visit the factory during manufacture of any or all items covered by this specification, for inspection of material or witness of tests. Accordingly, the manufacturer shall give ...EDC testing schedule.

7.1 Design Tests:

- Polymer long rod insulator shall be tested for the design tests given in Table (5).

Table (5)

No.	Test Description	IEC	Clause No.
1	Test on interfaces and connection of end fittings		
1.1	Sudden load release	61109	10.3.1
1.2	Thermal Mechanical pre-stress	61109	10.3.2
1.3	Water immersion test	62217	9.2.6
1.4	Verification test	62217	9.2.7
1.5	Visual examination	62217	9.2.7.2
1.6	Steep front impulse voltage	62217	9.2.7.3
1.7	Dry power frequency voltage test	62217	9.2.7.4
2	Test on shed and housing material		
2.1	Hardness test	62217	9.3.1
2.2	Accelerated weathering test	62217	9.3.2
2.3	Tracking and erosion test	62217	9.3.3
2.4	Flammability test	62217	9.3.4
3	Test on core material		
3.1	Dye penetration test	62217	9.4.1
3.2	Water diffusion test	62217	9.4.2
4	Assembled core load time test		
4.1	Determination of the average failing load of the core of the assembled insulator	61109	10.4.2.1
4.2	Verification of the 96 hours withstand load test	61109	10.4.2.2
5	Hydrophobicity	62073	6
6	Acid resistance of rod	62039	3.8



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7.2 Type Tests:

- Polymer long rod insulator shall be tested for the electrical tests given in Table (6).

Table (6)

No	Test Description	Test Standard
1	Dry lightning impulse withstand voltage test	IEC 61109 cl. 11.1 IEC 60383-2
2	Wet power frequency withstand voltage test	IEC 61109 cl. 11.1 IEC 60383-2

- Polymer long rod insulator shall be tested for the mechanical test given in Table (7).

Table (7)

No	Test Description	Test Standard
1	Damage limit proof test of the tightness of the interface between end fittings and insulator housing.	IEC 61109 cl. 11.2

7.3 Sample Tests:

- Polymer tension-type insulator shall be tested as per Clause 12 of IEC 61109 for the following sample tests. The sample size shall be as given in Table (8):
 - a) Verification of dimensions. (E1 + E2)
 - b) Verification of the locking system. (E2)
 - c) Verification of the tightness of the interface between end fittings and insulator housing. (E2)
 - d) Verification of the Specified Mechanical Load (SML). (E1)
 - e) Galvanizing Test. (E2)



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Table (8)

Lot size N	Sample size	
	E1	E2
$N \leq 300$	Subject to agreement	
$300 < N \leq 2\,000$	4	3
$2\,000 < N \leq 5\,000$	8	4
$5\,000 < N \leq 10\,000$	12	6

7.4 Routine Tests:

- Polymer long rod tension–type insulator shall be tested as per Clause 13 of IEC 61109 for the following routine tests:
 - a) Mechanical routine test.
 - b) Visual examination.

8. GUARANTEE

- The supplier guarantee the supplied materials against all defects arising out of faulty design or workmanship, or of defective material for a period of five years at least from date of delivery.
- Any defective parts of poor-quality or poor-assembly materials that are detected during the guarantee period should be replaced or repaired as soon as possible and free of charge.

9. SUBMITTALS

9.1 Submittals required with tender:

- The supplier shall complete and return one copy of the attached Data Schedule.
- Technical data and original catalogues shall be submitted with the offer.



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- One specimen of the offered insulator shall be submitted with the offer for evaluation prior to issuance of the purchase order. The specimen should correspond in all respects with the material they intend to supply. Dimensional verification, indelible markings and finishing shall be checked.
- All necessary drawings including layout, dimensions, should be submitted with the offer. The drawing should include but not be limited to the following information:
Nuts and spring washer shall be hot dip galvanized.
- A valid approval certificate issued by Egyptian Electricity Holding Company.
- Copy of type test reports.
- List of deviations & clauses to which exception is taken (if any).

9.2 Submittals required following award of contract:

- Details of manufacturing and testing schedules
- Routine test reports
- The supplier should provide all the instructions of use/storing in (Arabic and English).

10. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.



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Technical Data Schedule

NO	ITEMS	Vendor Specifications
1	Nominal system voltage	kV
2	Highest system voltage	kV
3	Frequency	 Hz
4	Lightning impulse withstand voltage 1.2/50 μ s (LIWL) kV peak	kV
5	Minimum specific leakage path based on highest voltage	 mm/kV
6	Minimum breaking load	kN
7	Dry power frequency flashover voltage	kV
8	Wet power frequency flashover voltage	kV
9	Dry power frequency withstand voltage	kV
10	Wet power frequency withstand voltage	kV
11	Diameter of shed (large / small)	mm /mm
12	Diameter of core with housing	mm
13	Minimum thickness of housing	 mm
14	Withstand bird biting	Yes/No
15	Color of insulation	
16	Total length	mm
17	Weight of insulator	kg
18	Marking as per clause 6	Yes/No
19	Corona extinction voltage	kV
20	Type of material components: - Rod - Weather sheds - Bottom Fitting - Top Fitting	
21	Guarantee period	year

We guarantee the data given above for the offered insulator toEDC.

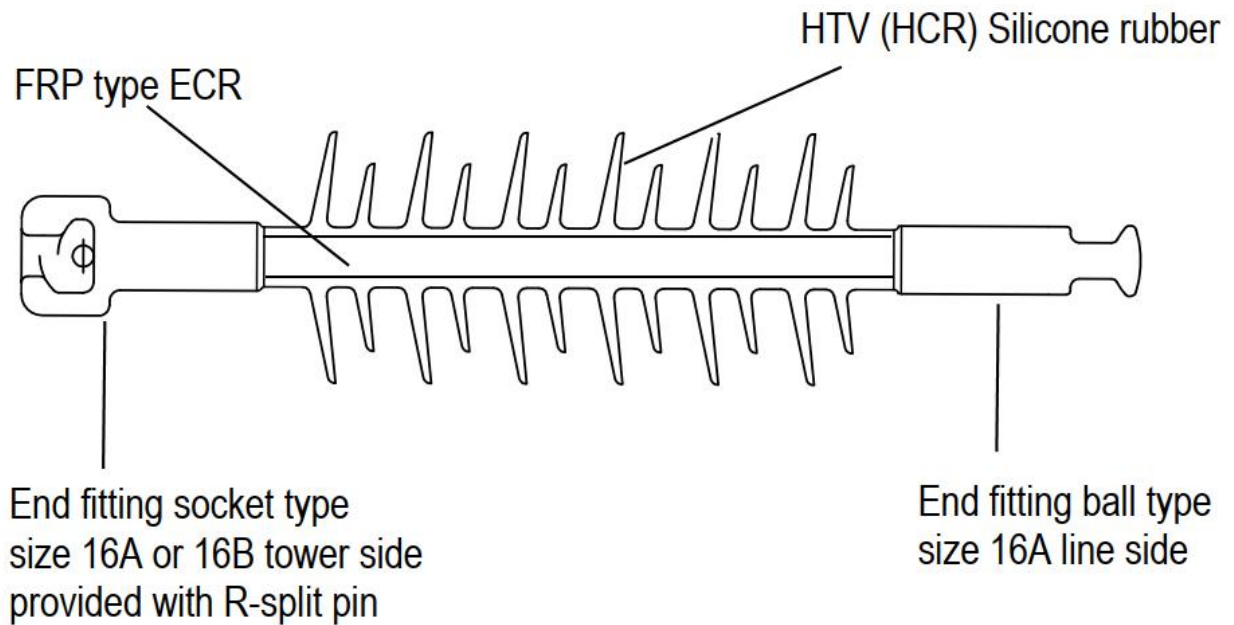
Signature: Date:

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Long-rod Tension-type Polymer Insulator 12 and 24 kV



- **c** is the minimum distance between adjacent sheds of the same diameter.
- Negative tolerance shall not be applicable to creepage distance.
- The tolerance on dimension **d** (in millimeters) is:
 $\pm (0.04d + 1.5) \text{ mm}$ when $d \leq 300 \text{ mm}$
 $\pm (0.025d + 6) \text{ mm}$ when $d > 300 \text{ mm}$ with a maximum tolerance of $\pm 50 \text{ mm}$,

